

HUOYUAN TAN

Robotics Algorithm Engineer

Autonomous Mobile Robotics | Robot Perception | Embodied AI



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ABOUT ME

Robotics algorithm engineer with **5+ years** of experience developing and deploying **autonomous mobile robot systems** for ports, factories, and logistics. Hands-on work in **perception, SLAM, motion planning, control, multi-sensor fusion**, and ROS-based robot system implementation in **C++ and Python**. Recently completed an **M.Sc. in Computer Science and Engineering** at the University of Gothenburg (2026), with a thesis on **Vision-Language-Action (VLA)** assessment and failure recovery. Seeking **Robotics Algorithm Engineer** roles in autonomous mobile robotics and embodied AI.

TECHNICAL SKILLS

Robot Perception:	2D/3D detection and tracking, semantic segmentation, camera-LiDAR calibration, multi-camera calibration, sensor fusion.
Localization & SLAM:	LiDAR graph-based SLAM, Cartographer, ORB-SLAM3, ICP/NDT, LiDAR-IMU localization, EKF/UKF state estimation.
Planning & Control:	A*, Hybrid-A*, TEB, trajectory optimization, PID, MPC, navigation and decision-making for mobile robots.
Robot Learning & AI:	PyTorch, Hugging Face, deep learning, reinforcement learning, VLA models, LLMs, MuJoCo simulation, failure detection/recovery.
Programming & Tools:	C++, Python, ROS/ROS2, Linux, CMake, Docker, Git.
Sensors & Interfaces:	LiDAR, RGB/RGB-D cameras, GNSS/RTK, IMU, CAN, Modbus, MQTT.

PROFESSIONAL EXPERIENCE

Robotics Algorithm Engineer | [Xingyun Intelligence \(Shenzhen\) Technology Co., Ltd.](#) 2017.02 – 2022.10
Shenzhen, China

- Developed and deployed autonomous forklift, AGV, and intelligent guided vehicle / autonomous port truck systems from prototype validation to real-world operation in ports and factories.
- Implemented **perception, SLAM, planning, and control** modules in **C++/ROS**, using cameras, LiDAR, GNSS/RTK, and IMU for multi-sensor fusion and reliable navigation.
- Supported field commissioning by diagnosing localization, navigation, and control issues through log analysis, sensor calibration, parameter tuning, and cross-team debugging.

SELECTED PROJECTS

Industrial Autonomous Mobile Robot Systems — AGV / Autonomous Forklift / Port Truck 2018 – 2022
video: [@yunxing-port/videos](#)

- Developed multi-LiDAR calibration and sensor fusion for port-truck obstacle avoidance/localization; implemented LiDAR-camera calibration and **MPC-based** motion planning for autonomous forklifts, reaching **90–92%** loading/unloading success rates.
- Built LiDAR-IMU **SLAM** with loop closure for AGV mapping, localization, and map extension across indoor and outdoor factory environments.

VLA Robot Learning — Assessment & Failure Recovery (M.Sc. thesis) 2025 – 2026
code: github.com/HyuanTan/lerobot_far

- Developed assessment and failure-recovery methods for **Vision-Language-Action (VLA)** manipulation policies in remote deployment; validated in MuJoCo simulation and on the open-source SO-101 robot arm.

Indoor Activity Recognition with Pyroelectric Infrared Sensors 2015.06 – 2017.02

- Researched indoor human tracking and abnormal activity detection using Random Forests, HMMs, and One-Class SVMs; results published in *Sensors*.

EDUCATION

M.Sc. in Computer Science and Engineering | [University of Gothenburg](#) 2024.09 – 2026
Gothenburg, Sweden

- Thesis: Assessment & Failure Recovery in Remote Vision-Language-Action Deployment. Selected courses: Deep Machine Learning, Advanced Topics in Machine Learning, Design of AI Systems.

B.Eng. Medical Information Engineering | [Guangzhou University of Chinese Medicine](#) 2013.09 – 2017.06
Guangzhou, China

- Outstanding Graduate Award, 2017.

PUBLICATIONS & LANGUAGES

Publications: Two peer-reviewed papers in *Sensors*: *Simultaneous Indoor Tracking and Activity Recognition Using Pyroelectric Infrared Sensors* (2017, 3rd author) and *Abnormal Activity Detection Using Pyroelectric Infrared Sensors* (2016, 2nd author). **Languages:** English (fluent for everyday and academic communication); Mandarin Chinese (native); Cantonese (native).